



Backup and Restore

Agenda

- ▶ **Definition, Importance and Purpose**
- ▶ **Types of Backups**
 - ▶ **Advantages and Disadvantages**
 - ▶ **Recovery**
 - ▶ **Schedule**
- ▶ **Example**



Backup

Definition, Importance & Purpose

Backup

- ▶ A database backup is the process of creating a copy of a database's data and structure.
- ▶ These copies are stored separately and can be used to restore the database to a previous state, ensuring data integrity and business continuity.

Importance & Purpose

- ▶ Database backup provides a safeguard against:
 - ▶ Accidental data loss
 - ▶ Data corruption
 - ▶ Unforeseen disasters
 - ▶ Security against attacks

Importance & Purpose

- ▶ Database backup ensures the continuity of business operations.
- ▶ With databases often storing vital information, a backup plan minimizes risks for data loss and availability.

Importance & Purpose

- ▶ Backups help avoid severe operational disruptions and costly data recreation efforts.
- ▶ Regulatory requirements often mandate data protection measures, including regular backups for compliance purposes.

Importance & Purpose

- ▶ In some industries, organizations need to adhere to data protection laws, making database backups a legal or commercial necessity.



Types of backup

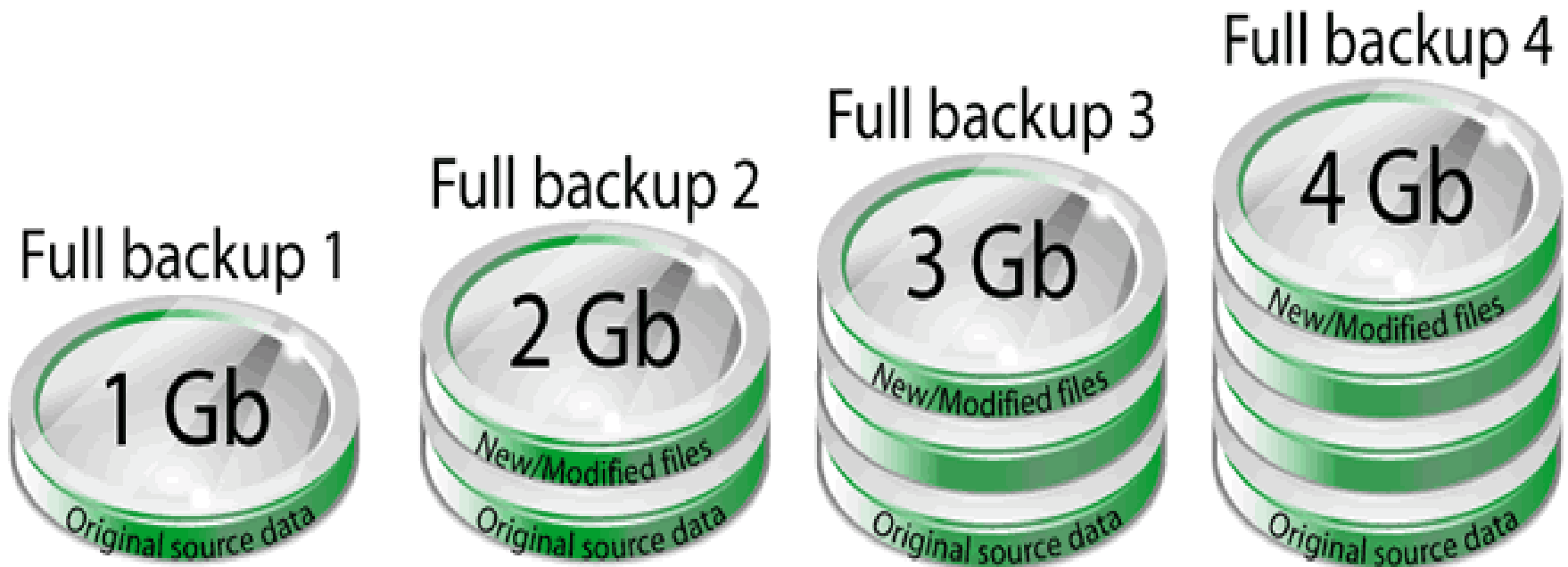
Types of Backups

- ▶ Following are the types of Backup:
 - ▶ Full Backup
 - ▶ Differential Backup
 - ▶ Incremental Backup
- ▶ A combination of the above types of backup makes a **backup schedule** for an efficient recovery method.

Full Backup

- ▶ This method copies the entire database and all its associated data files, configuration files, and necessary log portions.
- ▶ This method is not dependent on any previous backups.
- ▶ Complete copy of database at a point of time.

Full Backup - Schedule



Note: Full Backup will always back-up the entire source data. If you don't delete/exclude sources (only add/modify) it will always grow in size because it backs up everything.

Full Backup

► Pros:

- Minimal time is required to restore.
- Provides single point of recovery.
- Reliable and complete.

► Cons:

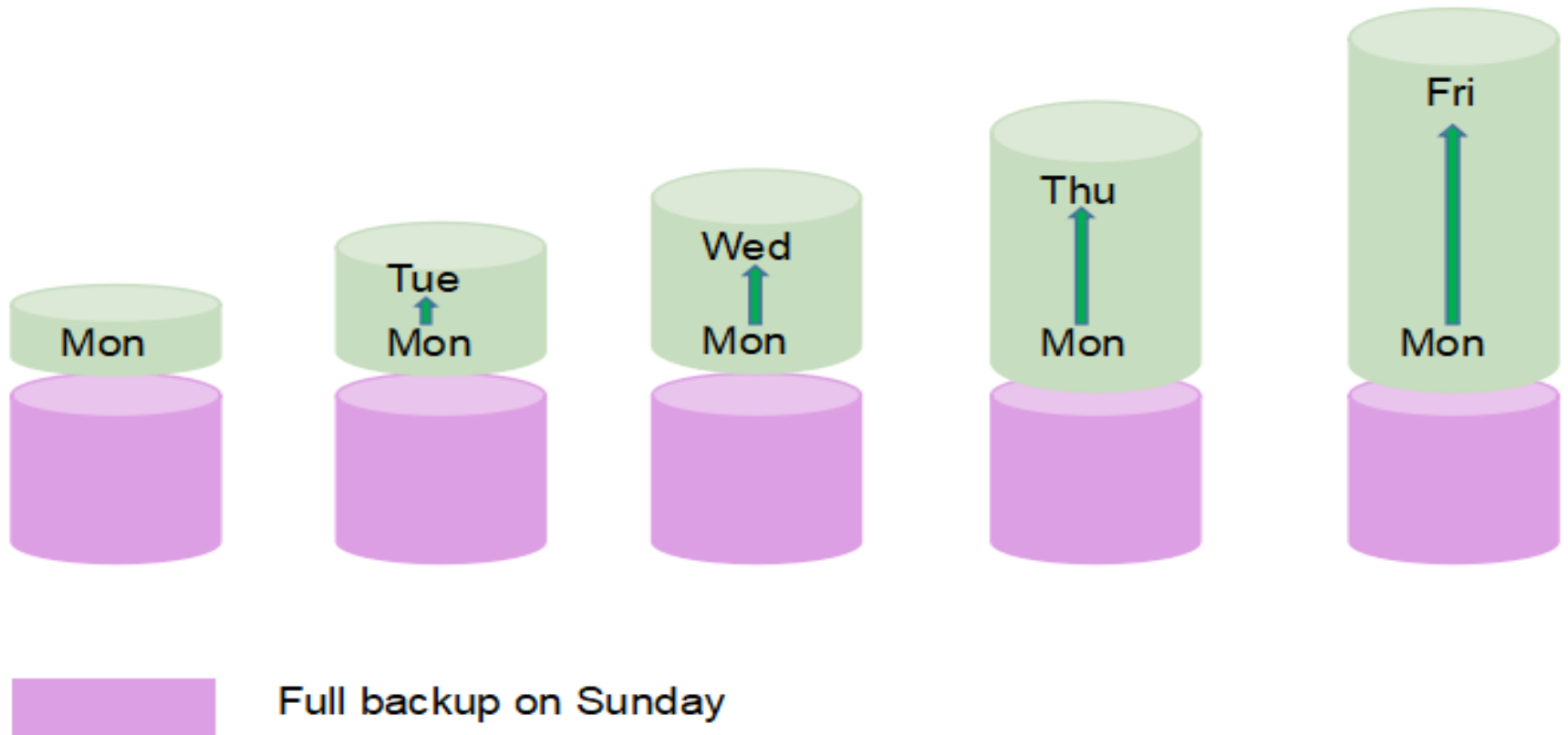
- Time-consuming to create.
- Requires the most storage space.
- Not efficient if done too frequently

Differential Backup

- ▶ A differential backup stores all changes made after the most recent full backup.
- ▶ It depends on the last full backup taken.
- ▶ It will grow until next full backup is taken.
- ▶ This is ideal for systems with low data modification rate and high reads.

Differential Backup - Schedule

Differential Backup



Differential Backup

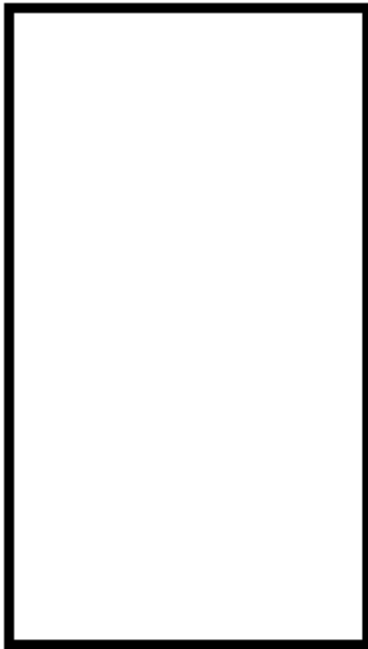
► Pros:

- Faster to create than a full backup
- Requires less storage space than full backup.
- Restoring only requires the last full backup and the latest differential backup.

► Cons:

- The backup size grows over time until the next full backup is performed.

Differential Backup - Recovery



FULL BACKUP



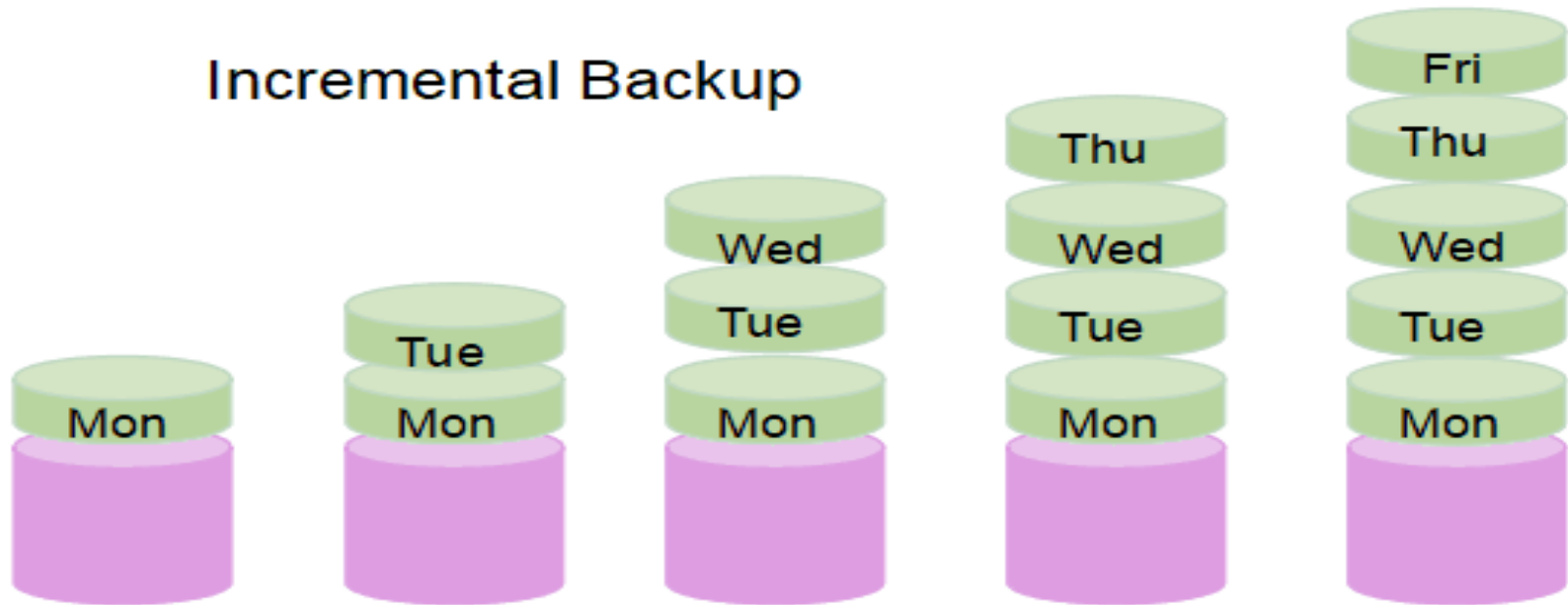
**LAST DIFFERENTIAL
BACKUP**

Incremental Backup

- ▶ An incremental backup captures only the new or modified data since the last full or incremental backup.
- ▶ It's the smallest and fastest type of backup.
- ▶ It is ideal for organizations with large datasets and large amount of changing data.

Incremental Backup - Schedule

Incremental Backup

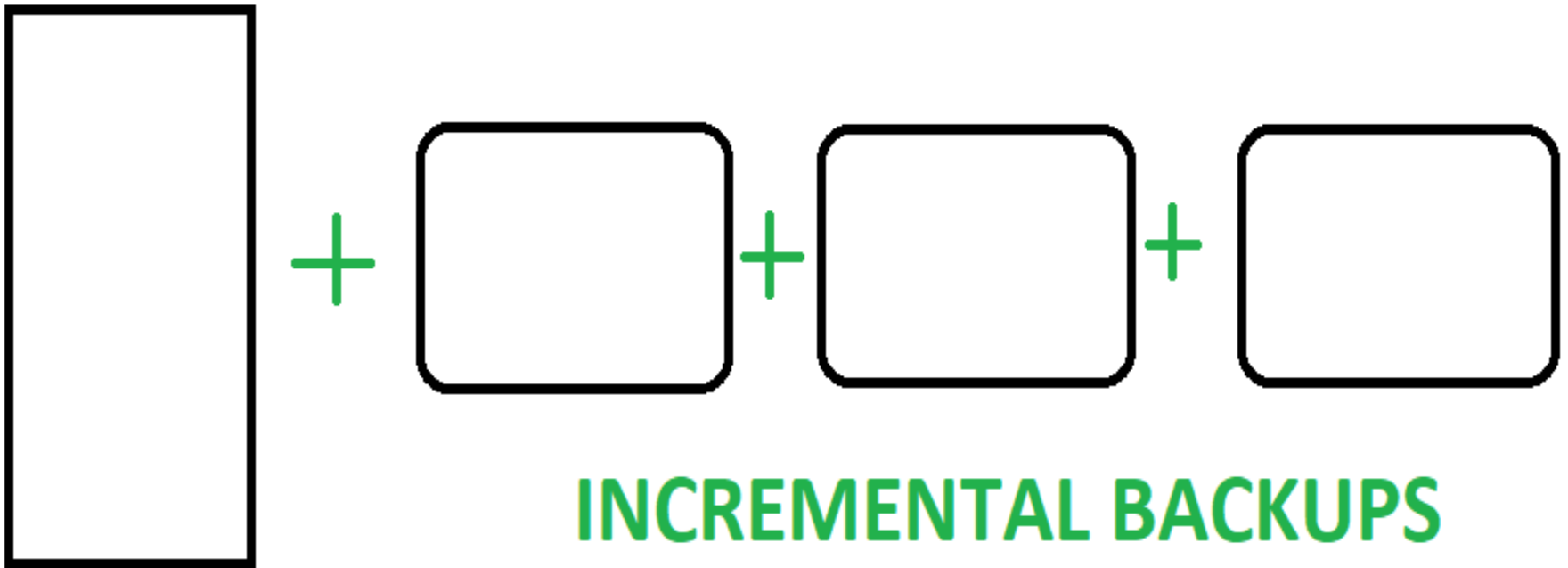


Full backup on Sunday

Incremental Backup

- ▶ **Pros:**
 - ▶ Fastest backup (copies only recent changes)
 - ▶ Requires least storage space
 - ▶ Suitable for systems with frequent backups
- ▶ **Cons:**
 - ▶ Slower restore — must apply multiple incremental backups in sequence
 - ▶ Complex management
 - ▶ High risk if one backup in the chain is corrupted

Incremental Backup - Recovery



FULL BACKUP

INCREMENTAL BACKUPS

Summary

FULL

INCREMENTAL

DIFFERENTIAL

Description:	copies the entire data set	full backup + changes since the previous backup	full backup + changes since the full backup
Backup time:	time-consuming	fast to back up	faster than a full backup but slower than an incremental
Recovery time:	fast recovery	slow recovery	faster than incremental but slower than full
Storage space:	requires a lot of storage space	requires less storage space	requires less storage space than a full backup, but more than an incremental
Bandwidth:	uses a lot of bandwidth	uses less bandwidth	uses less bandwidth than a full backup, but more than an incremental backup



Example

Example

- ▶ Consider the following Backup Schedule:
 - ▶ Every Sunday 9 am Full Backup
 - ▶ Mondays to Saturdays Incremental Backup at 9 am
- ▶ In case a disaster occurs at 1:55 pm on a **Thursday** resulting in loss of hardware (including storage), answer the following:
 - ▶ How much data (in terms of time) will be lost?
 - ▶ What would be the restoration steps?
 - ▶ Will the restoration be faster if the incremental backups are replaced with differential ones? Why?

Example

- ▶ Consider the following Backup Schedule:
 - ▶ Every Sunday 9 am Full Backup
 - ▶ Mondays to Saturdays Incremental Backup at 9 am
- ▶ Disaster occurs at 1:55 pm on a Thursday
- ▶ How much data (in terms of time) will be lost?
 - ▶ *4 hours and 55 minutes*

Example

- ▶ Consider the following Backup Schedule:
 - ▶ **Every Sunday 9 am Full Backup**
 - ▶ **Mondays to Saturdays Incremental Backup at 9 am**
- ▶ Disaster occurs at 1:55 pm on a **Thursday**
- ▶ What would be the restoration steps?
 - ▶ *Restore latest Full Backup taken on Sunday*
 - ▶ *Restore Incremental Backups taken on Monday, Tuesday, Wednesday & Thursday.*

Example

- ▶ Consider the following Backup Schedule:
 - ▶ Every Sunday 9 am Full Backup
 - ▶ Mondays to Saturdays Incremental Backup at 9 am
- ▶ Disaster occurs at 1:55 pm on a Thursday
- ▶ Will the restoration be faster if the incremental backups are replaced with differential ones? Why?
 - ▶ Yes
 - ▶ *The Restoration will require only 2 backups:*
 - ▶ *Full Backup taken on Sunday*
 - ▶ *Differential Backup taken on Thursday*



THANK YOU